

Line Search and Constrained Minimization

Optimization and Learning for Robot Control

RCI Lab

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Previously:

- Root finding
- Newton's method
- Minimization
- Regularization

Problem:

Newton step may overshoot.

Newton Step

Newton direction:

$$H(x_k)p_k = -\nabla f(x_k)$$

Update:

$$x_{k+1} = x_k + p_k$$

Issue:

Step may increase $f(x)$.

Line Search Idea

Instead use:

$$x_{k+1} = x_k + \alpha_k p_k$$

Choose $\alpha_k \in (0, 1]$.

Goal:

Ensure sufficient decrease.

Armijo Condition

First-order Taylor expansion:

$$f(x + \alpha p) \approx f(x) + \alpha \nabla f(x)^T p$$

Armijo rule:

$$f(x + \alpha p) \leq f(x) + c\alpha \nabla f(x)^T p$$

with:

$$0 < c < 1$$

Backtracking Algorithm

Initialize:

$$\alpha = 1$$

While Armijo not satisfied:

$$\alpha \leftarrow \beta \alpha$$

where:

$$0 < \beta < 1$$

Typical values:

$$c = 10^{-4}, \quad \beta = 0.5$$

Equality-Constrained Minimization

Problem:

$$\min_x f(x)$$

subject to:

$$c(x) = 0$$

where:

$$c : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Lagrangian

Define:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T c(x)$$

First-order conditions:

$$\nabla_x \mathcal{L} = \nabla f(x) + J_c(x)^T \lambda = 0$$

$$c(x) = 0$$

KKT System (Equality Case)

Linearize:

$$\begin{bmatrix} H & J_c^T \\ J_c & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} \nabla f + J_c^T \lambda \\ c(x) \end{bmatrix}$$

where:

$$H = \nabla_{xx}^2 \mathcal{L}$$

Newton update:

$$x \leftarrow x + \Delta x$$

$$\lambda \leftarrow \lambda + \Delta \lambda$$

Gauss–Newton Method

When:

$$f(x) = \frac{1}{2} \|r(x)\|^2$$

Gradient:

$$\nabla f = J_r^T r$$

Exact Hessian:

$$\nabla^2 f = J_r^T J_r + \sum_i r_i \nabla^2 r_i$$

Gauss–Newton approximation:

$$H \approx J_r^T J_r$$

Ignore second term.

Why Gauss–Newton?

Pros:

- No second derivatives
- Always positive semi-definite
- Cheaper per iteration

Con:

Slower than full Newton (more iterations).
Often better in wall-clock time.

Inequality Constraints

Problem:

$$\min_x f(x)$$

subject to:

$$c(x) \leq 0$$

KKT Conditions (Inequality)

Stationarity:

$$\nabla f(x) + J_c(x)^T \lambda = 0$$

Primal feasibility:

$$c(x) \leq 0$$

Dual feasibility:

$$\lambda \geq 0$$

Complementarity:

$$\lambda_i c_i(x) = 0$$

Key Takeaways

- Newton is local — needs globalization.
- Line search ensures descent.
- Equality constraints $\rightarrow$ Lagrange multipliers.
- Inequality constraints $\rightarrow$ KKT conditions.
- Gauss–Newton widely used in robotics.